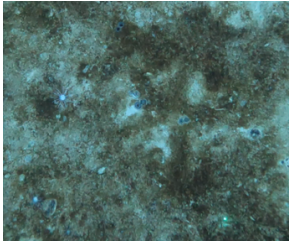


Supporting Information for:

Microhabitat patchiness structures benthic biodiversity in the Western Antarctic Peninsula

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Supplementary Table 1 - Description and characteristics of most abundant SIMPROF groups

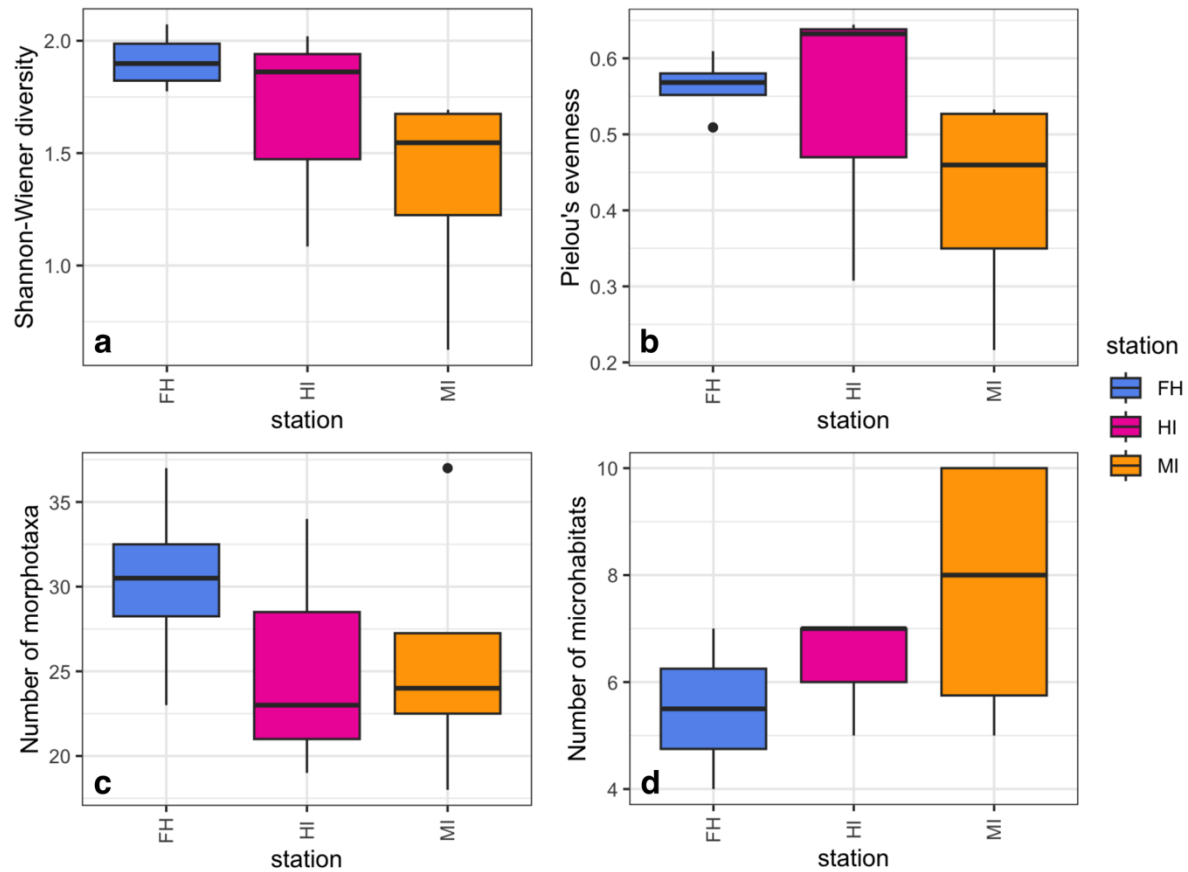
Group	Found in	Description (For most common morphotaxa, most probable identification is put between brackets)	Total n of taxa	Mean n of taxa per image	Key taxa (SIMPER)	Key image
e	MI: 0 HI: 22 FH: 0	Dominated by rocky substrate and large amounts of diatom mats. Faunal community composed mainly by <i>Odontaster validus</i> (pink_star_msp1), <i>Heterocucumis steineni</i> (blue_holoth_msp1), terebellid worms and <i>Doris kerguelensis</i> (white_bunny_msp1). Notable lack of <i>Margarella antarctica</i> (white_snail) and <i>Laternula elliptica</i> compared to the other groups. Groups d , f and g are very similar to e but have some rarer species within their images.	23	5.75	<u>Macroalgae:</u> filamentous_filiform_algae <u>Fauna:</u> pink_star_msp1 blue_holoth_msp1 terebellidae white_bunny_msp1	 TANGO2_ROV13_2_027.png (HI)
h	MI: 6 HI: 47 FH: 1	Typical soft bottom habitat. High abundance in <i>L. elliptica</i> and <i>Edwardsiella andrillae</i> , and just like group e , a distinct lack of <i>M. antarctica</i> .	20	3.23	Very little macroalgae, no dominant one <u>Fauna:</u> edwardsiella_msp4 laternula	 TANGO2_ROV14_7_16.png (HI)
i	MI: 6 HI: 4 FH: 1	Also soft bottom, but faunal community of group i is very different from group h . The most abundant morphotaxa is white_anemone_msp3 and <i>M. antarctica</i> is also more frequently present, but the total number of taxa in this group is very low (9).	9	2.2	white_anemone_msp3	 TANGO2_ROV19_1_045.png (FH)

j	MI: 7 HI: 4 FH: 0	Many different macroalgae and starfish morphotaxa.	18	3.36	<u>Macroalgae:</u> laminar_brown encrusting_algae desmarestia_msp2 <u>Fauna:</u> sun_star_msp2 encrusting_sponge	 TANGO2_ROV10_3_004.png (MI)
n	MI: 10 HI: 0 FH: 1	Images from group n all have some amount of <i>Plocamium hookeri</i> (branching_red_algae_msp1). The faunal community is quite cryptic, with some light_brown_star_msp15 and beige_bivalve_msp1, blending in with the sandy bottom seascape.	15	3.61	<u>Macroalgae:</u> red_or_brown_sheet_algae_msp0 filamentous_green_algae <u>Fauna:</u> beige_bivalve_msp1 light_brown_star_msp15 small_white_worm_msp2	 TANGO2_ROV9_1_028.png (MI)
s & t	(s) MI: 15 HI: 0 FH: 0 (t) MI: 29 HI: 0 FH: 0	Groups s and t are very similar in their macroalgae community composition. The dominant algae morphotaxa is <i>P. hookeri</i> , in much higher density than group n . The difference between the two groups is the faunal composition living within the dense bushes.	12 (s) & 22 (t)	3.44 (s) & 2.63 (t)	<u>Macroalgae:</u> branching_red_algae_msp1 <u>Fauna:</u> large_yellow_msp3 skinny_white_star_msp12 white_snail small_yellow_star_msp6 skinny_white_star	 TANGO2_ROV6_1_028.png (MI)
x	MI: 3 HI: 15 FH: 38	Very diverse macroalgae bushes with sponges (<i>Dendrilla antarctica</i>) and anemones.	44	6.69	<u>Macroalgae:</u> branching_red_algae_msp2 <u>Fauna:</u> spiky_yellow_msp7 orange anemone_msp1	 TANGO2_ROV14_2_046.png (HI)

ab	MI: 10 HI: 9 FH: 9	Very diverse macroalgae forest, although different from group j , with a high abundance of <i>Iridaea cordata</i> (red_sheet_msp1) and <i>Desmarestia antarctica</i> (desmarestia_msp1). A common feature with group x is the high abundance of <i>Dendrilla antarctica</i> .	26	3.8	<u>Macroalgae:</u> desmarestia_msp1 red_sheet_msp1 <u>Fauna:</u> finger_sponge_msp1 small_pink_worm_msp1	 TANGO2_ROV20_3_013.png (FH)
ac	MI: 0 HI: 0 FH: 11	Groups ab and ac cluster together due to their high abundance of <i>I. cordata</i> , but group ac 's faunal community is very distinct from ab . The most influential morphotaxa distinguishing this group from the others are <i>Sterechnus neumayeri</i> (urchin_msp1) and <i>Nacella concinna</i> . Main substrate type is rock or other consolidated types, covered by pink_encr_algae_msp1 (probably <i>Leptophytum coulmanicum</i>).	18	8	<u>Macroalgae:</u> red_sheet_msp1 pink_encr_algae <u>Fauna:</u> laternula urchin_msp1 nacella spiky_massive_msp9 parborlasia_corrugatus	 TANGO2_ROV16_2_026.png (FH)
ad	MI : 2 HI: 0 FH: 23	Similarly to group ab and ac , the abundant <i>I. cordata</i> is one of the defining morphotaxa of group ad . Some fauna in common with group ac , although less dominant (this is the case for <i>N. concinna</i> , spiky_massive_msp9, pink_encr_algae_msp1 and <i>L. elliptica</i>). The most contributing morphotaxa of this group to the differences with the other groups is the high abundance of the sponge morphotype white_massive_msp10.	30	6.69	<u>Macroalgae:</u> red_sheet_msp1 <u>Fauna:</u> white_massive_msp10 white_snail	 TANGO2_ROV21_1_065.png (FH)
ae	MI: 13 HI: 0 FH: 16	Images from this group distinguish themselves from the others by their high cover of <i>Himantothallus grandifolius</i> and high abundance of brittle stars.	41	7.06	<u>Macroalgae:</u> himantothallus_msp1 <u>Fauna:</u> brittle_msp1 encrusting_sponge Other stars	 TANGO2_ROV10_4_013.png (MI)

Supplementary Table 2 – Summary table of station biodiversity indexes

Station	Number of morphotaxa	Shannon-Wiener diversity	Pielou's evenness	Number of microhabitats
HI	45	2.326	0.611	17
MI	53	1.611	0.405	19
FH	50	2.117	0.541	12



Supplementary Figure 1 - Boxplots of biodiversity metrics of studied sites. (a) Shannon-Wiener diversity index. (b) Pielou's evenness index. (c) Number of different morphotaxa. (d) Number of different microhabitats

SIMPER Analysis results on microhabitat groupings:

Microhabitat

Similarity Percentages - Morphotaxa contributions

One-Way Analysis

Parameters

Resemblance: S17 Bray-Curtis similarity

Cut off for low contributions: 70.00%

Between Stations:

Group MI

Average similarity: 29.66

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
branching_red_algae_msp1	4.87	18.62	0.72	62.78	62.78
red_sheet_msp1	2.01	6.52	0.73	21.97	84.75

Group HI

Average similarity: 17.21

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
edwardsiella_msp4	0.97	3.71	0.41	21.58	21.58
laternula	1.06	3.63	0.35	21.1	42.68
filamentous_filiform_algae	1.99	3.26	0.29	18.93	61.61
red_sheet_msp1	1.28	2.24	0.37	13.03	74.63

Group FH

Average similarity: 34.07

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	3.94	12.07	0.94	35.42	35.42
branching_red_algae_msp2	2.69	6.64	0.65	19.48	54.91
white_snail	1.99	6.33	0.97	18.58	73.49

Dissimilarity matrix:

	MI	HI
MI		
HI	90.8	
FH	81.2	85.7

Between main SIMPROF groups (microhabitats):

Group ab

Average similarity: 56.13

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	8.79	51.57	3.89	91.89	91.89

Group ae

Average similarity: 38.06

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	2.34	11.9	1.83	31.28	31.28
white_snail	2.36	10.38	1.38	27.28	58.56
himantothallus_msp1	3.45	8.97	0.65	23.58	82.14

Group j

Average similarity: 27.52

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	1.86	19.05	1.27	69.2	69.2
desmarestia_msp2	2.36	7.05	0.55	25.62	94.83

Group i

Average similarity: 51.36

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
white_snail	1.73	48.77	2.75	94.96	94.96

Group e

Average similarity: 68.10

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
filamentous_filiform_algae	8.64	49.59	6	72.82	72.82

Group h

Average similarity: 40.81

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
laternula	2.73	23.76	1.26	58.23	58.23
edwardsiella_msp4	1.79	13.81	0.93	33.85	92.07

Group x

Average similarity: 46.14

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
branching_red_algae_msp2	5.92	28.93	2.46	62.7	62.7
white_snail	1.85	5.09	0.85	11.03	73.73

Group ad

Average similarity: 54.71

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	6.39	31.41	3.94	57.41	57.41
white_snail	2.88	13.22	2.69	24.16	81.57

Group ac

Average similarity: 60.44

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
red_sheet_msp1	6	19.94	3.14	32.99	32.99
laternula	4.01	14.14	1.57	23.4	56.39
urchin_msp1	2.85	11.95	3.82	19.77	76.16

Group n

Average similarity: 60.58

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
branching_red_algae_msp1	4.27	40.26	3.36	66.46	66.46
red_sheet_msp1	2.13	15.76	1.37	26.02	92.49

Group t

Average similarity: 82.65

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
branching_red_algae_msp1	9.71	77.26	9.84	93.48	93.48

Group s

Average similarity: 83.09

Species	Av.Abund	Av.Sim	Sim/SD	Contrib%	Cum.%
branching_red_algae_msp1	9.74	55.68	7.97	67.01	67.01

red_sheet_msp1

3.31 16.96 4.94 20.41 87.42

Dissimilarity matrix:

	e	h	i	j	n	s	t	x	ab	ac	ad
e											
h	97.2										
i	96.9	91.9									
j	95.4	93.4	94.4								
n	94.5	88.2	95.9	81.5							
s	94.1	93.1	85.3	84	50.8						
t	97.5	95.7	87.9	95.2	55.8	25.8					
x	92.4	85.6	84.3	89.7	84.6	79.5	88.3				
ab	93	93.8	95.2	78.9	78.1	74.8	92.4	82.2			
ac	93.3	77.9	87	84.9	80.2	75.5	91.1	68.6	66.3		
ad	89.6	87.1	79.6	81.3	73.1	62	81.8	71.1	58	56.2	
ae	92	90	79.2	81.4	77.4	70.1	83	74.9	78.6	71.3	62.5